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Contingency-resilient PMU placement using Fuzzy Logic and Discrete Artificial Bee Colony algorithm for comprehensive network observability

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ABSTRACT

In transforming traditional grid systems into intelligent and microgrid-based systems, the Phasor Measurement Units (PMUs) play a prominent role in enhancing monitoring, assessment, control, and protection by estimating voltage and current phasors correctly and quickly. However, since PMUs are not cost-effective, the decision to take the requisite number and their placement is a significant issue and needs to focus on a better approach. This study addresses the solution methodology to compute the optimal locations of a reduced PMU number to have complete network observability. A new hybrid optimization strategy based on the Fuzzy Logic (FL) and Discrete Artificial Bee Colony (DABC) algorithm has been suggested. The suggested method is titled FL-DABC algorithm, which has been applied to determining the optimal placement of PMUs in microgrid-based systems. The suggested approach tackles the problem of fully network-observable cost-effective PMU deployment. The effectiveness of the Algorithm is evaluated through simulations under various operating limitations, such as zero injection buses, a single PMU, and line outage scenarios. A topological observability analysis validates full network observability with the computed PMU placement. The study also includes branch PMU placement set computation and a comparative analysis with other techniques using standard IEEE test networks for simulations.

Introduction

Electricity has become the cornerstone for each society and industrial development. Any interruption in power supply, especially those leading to major blackouts, can hamper many important operations and create untimely inconvenience to consumers. This implies the requirement for an excellent monitoring and control system. Recently, the PMUs have attracted the attention of power engineers for real-time monitoring and retrieving the voltage and current phasor data continuously and at a very faster rate. These facilities, through PMUs, bring substantial improvement in wide-area operation and control, adaptive protection, higher level of security and reliability, computing correctly the state estimation, and accurate prediction with optimal forecasting [1]. In addition, detection and analysis of the events leading to major blackouts are possible through wide-area measurement systems with PMUs and existing measurement devices. A good monitoring system performs three important activities, including network topology determination, observability, and state estimation evaluation in a power system. Thus, there is a major planning concern for optimal PMUs installation to provide minimal cost design while ensuring full power system observability under different practical constraints and contingencies [2,3]. This study motivates considering this issue to formulate an adequate and efficient methodology to provide an optimal solution.

The real-time measurements through PMUs ensure the correct detection of any abnormal operation. Based on that, corrective actions can be taken to enhance the overall system's efficiency. The important factors that need to be focused on during the implementation stage are that PMUs are a very costly investment, and it is not mandatory to implement each bus PMU to make the system fully observable. That's why instead of installing on every bus, PMUs can be installed on selected buses in an optimized way to make the system observable [1]. Considering this issue, the optimal PMU placement problem (OPP) can be formulated as the problem of computing the minimum required number of PMUs and their corresponding installation bus locations such that the

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overall network becomes completely observable [4]. The complete system observability is assessed by knowing all the bus voltage and current phasor. Normally, two popular types of observability analysis methods have been used extensively to date. These are topological observability and numerical observability. This requires an optimization algorithm for finding an optimal solution during the planning stage to provide a fully visual system. The OPP problem is a combinatorial optimization problem with multiple optimal solutions [5,6]. Different objective functions, contingencies, constraints, and installation schemes are used in the OPP formulation. This study is researched to formulate a novel approach as a solution to the above problem for computing the minimum number of PMUs with the appropriate bus to install and getting the fully observable system even under different contingencies conditions.

This problem can be solved by two approaches, i.e., 1) classical or conventional approach, 2) non-classical or non-conventional approach [1,6]. The classical approaches are of two types iterative mathematical search and exhaustive search. Greedy algorithms and linear and nonlinear programs are the major methods applied under the iterative mathematical search for the OPP problem [7–10]. Graph theory and binary search are the methods that come under exhaustive search applied for the optimal PMUs placement. Heuristic, meta-heuristic, and machine learning techniques are used for the optimal location of PMUs and come under non-classical techniques. Many of the main heuristic methods, such as Monto Carlo simulation, Fuzzy decision-making, and unified algorithms, have been exploited for the OPP by many authors in recent times [1]. Among the metaheuristic approaches are Particle Swarm Optimization (PSO) [11–13], Differential Evolution (DE) [14,15], Genetic Algorithm [16,17], Simulated Annealing [18,19], and Ant Colony [20,21] are the major methods used for PMU placement problems. However, despite the many efficient methods suggested by many authors, it is found that the optimal placement of PMUs is still an open area of research as the complexity of transmission and distribution networks increases exponentially in smart and micro-grid scenarios.

Linear programming methods are failed to handle the nonlinear constraints. Constraints handling difficulty are the major limitation of these methods. Nonlinear programming complexity is comparatively high to handle large systems and more constraints-related problems. Heuristic and metaheuristic approaches have limitations related to computational time, and their efficiency depends on their system parameter setting. System topology, network model, operational constraints, different operational scenarios, and integration of DGs, Electric Vehicles (EVs), and energy storage systems (ESSs) affect all the method's performance. They must be improved further to handle these changes.

Moreover, many factors must be focused on to improve these methods, like handling capability of nonlinearity, data requirement, precision and accuracy, efficiency, and considering model uncertainty. From this perspective, evolutionary programming approaches are better than classical techniques because the solutions can be obtained quickly, and formulation is simple, even for large systems. Secondly, hybrid approaches make the associated limitations less in Comparison. Looking at this possibility, this study investigates further to formulate a method for the optimal PMU locations by proposing a hybrid approach comprised of fuzzy and discrete metaheuristic approaches.

Numerous works have been reported using different heuristic optimization algorithms [11–21]. The problem has been treated as a continuous optimization problem in these works. Very few binary optimization algorithms have been proposed for this problem. They are binary PSO (BPSO) [11,40] and modified binary PSO (MBPSO) techniques [9]. In [11], it is not possible to obtain a minimum optimal number of PMUs without the inclusion of conventional measurements. It is due to the absence of all the rules related to the zero injection bus in the observability analysis procedure. Also, the case of contingency conditions or the effect of branch PMUs is not considered. In [9], all those rules about zero injection bus are included, and the minimum has been obtained under normal and contingency conditions. However, the case of branch PMU placement is not considered. Apart from that, many of the associated limitations need these techniques a further improvement or go for a hybrid approach. These techniques do not allow for an assessment of the performance of redesigns, have a high risk of identifying low-priority issues, and are hard to incorporate in the modeling, the technical design constraints in the problem formulation and solution. This motivates a proposal of a hybrid approach to reduce the associated limitations in this study.

This work suggests a fuzzy modified DABC (FMDABC) optimization technique for optimal placement with fewer PMUs. The FMDABC is a binary optimization algorithm that is an enhanced version of the discrete ABC (DABC) algorithm, as suggested in [22,23,28]. The simulation test results indicate that the DABC algorithm fails to solve the PMU placement problem for different system conditions. The possible reasons for the failure of the DABC algorithm are a lack of efficient exploration in different directions of search spaces and local minimum trapping problems. The local search step used in the DABC algorithm must be efficient enough to find better solutions in any particular search direction in both the exploration and exploitation stage of searching. Also, the problem of the local minima trap can be avoided by incorporating random solution vectors in the population by which unexplored directions of search space can be utilized. Although this measure has been adopted in [28], frequent insertion of new solutions in every iteration leads to insufficient search in any particular direction because of losing some of the search directions that may be nearer to the optimal solution. To mitigate the above demerits of the DABC algorithm, three modifications are proposed in this work. They are (1) the introduction of a local search technique specifically suitable for the OPP problem, (2) the use of a fuzzy mutation probability for the new local search technique, and (3) the inclusion of stall limits in the Algorithm for not losing a feasible search direction. Both the DABC and FMDABC algorithms are applied on two standard benchmark functions of the uncapacitated facility location (UFL) problem as in [28] and compared in terms of various performance indices, proving the efficacy of the FMDABC algorithm. The major contribution of the research carried out in this study on the optimal placement of PMUs are as follows:

- Many operating contingencies are considered in the OPP formulation, like the effect of ZIBs, single PMU loss contingency, and single branch contingency outage. Additionally, complete observability under the placement of branch PMUs was investigated.
- Three modifications are suggested to enhance the searching capability of the approach in both the exploration and exploitation stage. (1) a swap operation in the local search step is proposed, which will change the number of 0 s and 1 s in the solution vector, and (2) local search mutation on the population is obtained using a fuzzy logic method; as a result, which the rate of execution of efficient local search is revamped consistently in every iteration, (3) introducing new random solution vectors into the population from time to time by which the probability of getting a better solution increases.
- A hybrid approach is formulated based on the DABC algorithm and FL for fuzzy mutation for the optimal location of the PMUs at the critical and sensitive buses.
- Simulation results with several IEEE test systems and comparative analysis with other suggested methods by various researchers are presented.
- A critical discussion of the best findings from the result with all its future scope of extension is presented to provide in-depth knowledge of the problem undertaken.

The remaining part of the presentation is structured as follows. To understand the study's objective, a brief introduction is presented on power system observability analysis in Section 2. The formulation of the problem with objectives, system constraints, and index terms is detailed with mathematical representation and discussed in Section 3. In Section 4, the suggested FMDABC approach is presented and explained with the formulation and process of finding the optimal solution. Comparative simulated results are presented in Section 5 under various contingency conditions to justify the efficacy of the novel suggested approach. The critical findings and future scope are enumerated in Section 6. Lastly, in Section 7, the manuscript concludes with concluding remarks on the entire study for finding an algorithm for optimal and minimum PMU placement in sensitive and critical buses.

Network observability analysis based on received pmu data

Observability, as understood from the control theory point of view, is a measure to infer the system's internal states to fully monitor the operational dynamics from the output response or knowledge of the system [24]. For any system, it is indispensable to get knowledge continuously, and then it can be said that the system is fully monitored or observable. Similarly, a power network can be considered fully observable subject to measurements from power flow analysis or load flow analysis; all other current and voltage phasors can be computed by state estimation. Observability analysis is an important task that needs to be evaluated while choosing the PMUs placement in power networks to continuously observe the states and their changes with changes in operating conditions through voltage and current measurements. In state estimation, observability analysis is considered one of the important areas to focus on in power networks, and PMU placement plays a crucial role in that. So, observability is crucial in installing bus PMUs [1]. Numerical observability analysis and topological observability analysis are the two main methods to compute the status of observability.

Numerical observability analysis

A network of N buses can be considered numerically observable subjected to the rank of the measurement function matrix obtained from the power system state estimation model is $2N$-1. For large-scale systems, the rank calculation for the numerical observability method may be computationally intensive and time-consuming each time a new PMU placement scheme is examined. Due to this issue, numerical observability analysis is avoided in this study as it requires intelligent methods to handle the associated limitations effectively. However, the solutions received by topological observability must be tested to ensure through numerical observability analysis before the final decision to place PMUs at their optimal positions. The model can be framed and mathematically expressed as in Eq.(1) [25,26]:

$$Z = HX + e \tag{1}$$

where Z denotes the measurement vector containing all measured parameters of voltages and currents. The dimension is the $M\,x1$ matrix. M is the number of available measurements. The Xdenotes the state vector containing N bus voltages and phase angles; the total state variable number is computed as *(2N-1)*. The $'e'$ is the measurement error vector. The$H$matrix's *(Mx(2N-1))* dimension signifies the state's and measurement vectors' relationship. The rank of the H matrix needs to be *(2N-1)* to indicate that the system is fully observable.

Topological observability analysis

This study focuses on topological observability analysis, which is much less computationally intensive than numerical observability analysis. In the case of topological observability analysis, a network consisting of N and M number of buses and branches, respectively, can be represented as a graph of N nodes and M edges. In this method, a spanning tree of full rank is designed to assess the status of full observability [24,26]. It is ensured by examining all system buses to be an observable minimum for once through the measurement scheme with a set of topological observability rules.

Rule 1: The rule of direct measurements states that in the case of a PMU integrated into the network on a bus is capable of measuring not only the voltage phasor of that bus but also the current phasor of all lines joined to that bus [24].

Rule 2: The rule of pseudo measurements states that the knowledge of the measured voltage phasor of both terminal buses of a branch makes that branch's current observable by calculating the current phasor of that concerned branch [9].

Rule 3: The knowledge of the bus voltage phasor of any terminal bus and that branch's current phasor makes the other terminal bus voltage observable [24,26].

Rule 4: A ZIB or any one of its neighboring buses can become observable by applying the following rules judiciously: [9,24]:

1 When the bus voltage phasor of all the neighboring buses to the ZIB is identified and accepts the ZIB itself, it is possible to determine applying Kirchhoff Current Law (KCL) to that bus and solving the KCL equations.
2 Similarly, in the case of the group of ZIBs, having the ZIB and all its neighboring buses, the voltage phasor of all the buses except anyone is unknown. It can be computed by solving the KCL equations of those ZI buses.
3 In the case of a group of ZIBs, which are interconnected, if the unknown bus voltage phasor number is equal to or one less than the ZIBs number, those unobserved bus voltage phasor is possible to calculate, making them observable.

Formulation of pmu placement problem

The optimal PMU placement in a large power system network can be interpreted as a single-objective or multi-objective problem where the system designers have to choose the optimal locations for PMU installation with the minimum number and full observability.

Basic PMU placement problem

For a network consisting of N nodes, the optimal and reduced number PMUs in an OPP is formulated and represented mathematically as follows in Eqs. (2) and (3).

$$\min \sum_{i=1}^{N} c_i x_i \tag{2}$$

Subject to

$$Obs(X) \geq b \tag{3}$$

where **X** denotes a binary solution vector. This reflects as a solution to the decision on placement of PMUs at different bus positions, i.e.,

$$x_i = \begin{cases} 1 & \text{if at bus i PMU is integrated} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

where c_i Represents the PMU cost placed on the i^{th}bus. It is considered to be equal for all PMUs. The '**b**' is an identity vector of size(Nx1) used in the problem formulation to ensure the observability of each bus at least once. The *"Obs"* function of the placement scheme X determines the number of times each bus is observable. In case of full observability with the normal operating condition and not considering zero injection effect, Eq. (3) can be described as:

$$AX \geq b \tag{5}$$

where **A** denotes the system connectivity matrix. It is having the connection information between various buses in the network whose elements are computed according to the logic presented in Eq. (6):

$$a_{ij} = \begin{cases} 1 & \textit{if bus i and j are integrated} \\ 1 & \textit{if } i = j \\ 0 & \textit{otherwise} \end{cases} \tag{6}$$

If the ZI effect is considered in the optimal position of PMUs, the observability analysis for a given solution vector is carried out as follows:

Step-1: The network connectivity matrix is formulated using Eq. (5) from the system connectivity information.
Step-2: The buses which can become observable with the application of *Rules* 1 and 2 (as presented in Section 2.2) are determined. All the observable buses can be identified from a vector **OV** resulting from multiplying the system connectivity matrix with the given solution vector. The positions containing any value other than 0 are considered observable buses.
Step-3: This step is only applied when the effect of the ZIBs is considered in the optimal PMUs problem formulation. The buses that can become observable with *Rule* 3 (Section 2.2) are determined. This step is recursively applied until no more buses become observable. At the end of this step, the observable bus positions in vector **OV** are set as 1 s.
Step-4: A constraint satisfaction value **CV** is assigned as 1 if all the elements of **OV** are non-zero values, indicating that the system is completely observable, else it is assigned a 0 value.

For the assignment of **CV** in case of PMU and line outage conditions, observability testing is carried out for each outage under consideration in the placement set. Only if it is possible to maintain the **CV** as 1 for all the outages considered is the final constraint value set to 1, indicating that the placement set can anticipate any outage circumstances and provide full observability.

Formulation of branch PMU placement problem

The branch PMU in OPP is discussed in this section. A combination matrix **C,** which stores all possible combinations of channel assignments when PMUs are integrated at different buses, is constructed. The number of rows contributed by a PMU-placed bus to the **C** matrix can be determined as follows [27]:

$$rows_i = \begin{cases} \dfrac{N_i}{N_{limit}(N_t - N_{limit})} & \textit{if } N_{limit} \le N_t \\ 1 & \textit{if } N_{limit} > N_t \end{cases} \tag{7}$$

where, N_i indicates the total number of branches connected to the i^{th}bus and N_{limit} refers to the maximum number of channel limits (2 for branch PMUs). The size of the **C** matrix now becomes$((\sum_i rows) \text{ x } N)$. A combination of possible channel assignments stored in the rows of the **C** matrix is such that only that column position indicates the bus number, and the column position indicates the adjacent bus number connecting these two buses through a branch to which a channel is assigned 1 s in the matrix. In this case, the ones in the binary solution vector denote the row positions representing a specific PMU bus and the corresponding channel assignment strategy. The major objective in this problem is to reduce the PMUs number in the placement scheme by which all the buses must become observable at least once. This can be expressed as:

$$\min \sum_{i=1}^{s} x_i \tag{8}$$

$$\text{Subject to } C^T X \ge b \textit{ and } G_j = 1 \textit{ for } j = 1....N \tag{9}$$

Where s is calculated as $\sum_i rows$ and **b** is a unit vector of dimension s. The parameter G_j denotes the number of PMUs placed on any bus and must not exceed 1. Each row in the combination matrix is selected to represent a possible PMU placement channel assignment strategy.

But, with this formulation, the solution may contain multiple PMUs with different channel assignment strategies at the same location. This also implies that a PMU should be placed on that bus with more than one number of channels, or two PMUs must be placed on the same bus. The effect of ZI is not included in the problem formulation because of the limited number of measurements available from the PMUs placement strategy, which results in very less measurement redundancy at each bus. Further, because of the consideration of conventional measurements in the system, any error in the measurement values might go undetected and cannot be tested due to the lack of a good set of measurements from branch PMUs. For example, in placing PMUs with multiple channels, the corresponding optimal placement set gets some of the buses observable multiple times, even without considering the effect of conventional measurements. Any error in those measured values can be easily detected because of the availability of those measurement values through another path. Depending upon the requirement and investment criteria of the utilities, any of the above placement strategies can be adopted.

Modified DABC algorithm

The ABC is a population-based meta-heuristic optimization technique proposed by Karaboga and Basturk [11,29]. It is modified in [28] specifically for discrete optimization problems.

Continuous version of ABC algorithm

The ABC algorithm is inspired by its basic searching principles formulated by the intelligent foraging behavior of the honeybee swarm. The Algorithm classifies all the foraging bees according to their behavior and responsibilities into three vital categories, i.e., employed bees, onlooker bees, and scout bees. All the bees collectively organize themselves to maximize the nectar amount, the energy source of bees, at the food stores in their hives through appropriate division of labor. The employed bees are the ones that exploit food sources by bringing them to their hives and visiting those food locations that were previously visited by the other employed bees. The onlooker bees waiting in their hives receive information about the explored food resources shared by the employed bees in the form of their dancing after their return to the hives. Afterward, the onlooker bees select to visit the food resources according to the quality of those recognized food resources from the duration of their dancing. The scout bees exploit new food sources in a random direction. The scout and onlooker bees are referred to as unemployed bees and get converted into employed bees after discovering a new food source during their search.

In the optimization process for a D-dimensional search space, the position of a food source of the D^{th} dimension is considered a possible solution. The nectar amount at the food source is formulated as the objective function value or fitness associated with that solution. The number of employed bees (Nb) equals the number of food sources (SN). It is because it is assumed that for every food source, only one employed bee [30]. The searching strategy of the basic ABC can be presented in four steps as follows:

Step-1: Initialization phase

In this phase, a population of SN number of solutions is randomly generated. The objective function evaluates the corresponding fitness values for selecting a better comparative solution. An individual 'i^{th}' solution vector in the population is represented by:

$$X_i = \{X_{i,1}, X_{i,2}, X_{i,3} .....X_{i,d}, ...X_{i,D}\} \quad \textit{where } i = 1, ..., SN \tag{10}$$

and all the individuals in the population of employed bees are represented by:

$$X = \{X_1, X_2, ...X_i, X_{i+1}......X_{SN}\} \tag{11}$$

The corresponding fitness values are calculated as follows:

$$fitness_i = \begin{cases} \dfrac{1}{1+f(X_i)} & iff\ (X_i)\ \geq 0 \\ 1 + abs(f(X_i)) & iff\ (X_i)\ < 0 \end{cases} \tag{12}$$

Step-2: Employed bees phase

In the employed bees phase, in each cycle '*t*,' a new solution vector is computed from the population of (*t*-1)th cycle as follows:

$$v^t_{i,d} = x^t_{i,d} + r^t_{i,d}\left(x^t_{i,d} - x^t_{k,d}\right) \tag{13}$$

Where $r^t_{i,d}$ is a randomly chosen scaling factor. The *k* is randomly selected between 1:Nb with $k \neq i$. This is followed by an evaluation of the fitness value, which, if found better than that of the corresponding employed bee food source location, replaces that food source location in the current cycle *t*.

Step-3: Onlooker phase

In this phase, a new set of solutions are generated, using Eq. (12), from the employed bees' food source locations. The probability value is denoted as p_i by which an onlooker bee chooses a food source and is computed as [29]:

$$P_i = \frac{fitness_i}{\sum_{j=1}^{SN} fitness_j} \tag{14}$$

The new solutions computed in the onlooker phase replace the previous solutions based on better fitness values.

Step-4: Scout phase

In this phase, the abandoned food sources by the employed bees are replaced by scout bees with a new computed random solution.

The discrete form of the ABC algorithm

A DABC algorithm proposed in [28] differs from the continuous one due to the improved binary mutation operator in the employed bee and onlooker phases. The concept of dissimilarity between two binary solution vectors is adopted for changing the differential expression for mutation operation in Eq. (13). This, in turn, depends upon their similarity. The degree of similarity between two binary vectors depends upon several quantities calculated from those two solution vectors. It is also defined per Jaccard's coefficient of similarity proposed in [30].

$$Similarity\ \left(X_i, X_j\right) = \frac{M_{11}}{M_{01} + M_{10} + M_{11}} \tag{15}$$

where M_{11}stores the total number of positions where both X_i, X_J have ones, M_{01}stores the total number of positions for which $X_i$has a 0 and X_j has a 1, and M_{10}stores the total number of positions for which X_i has a 1 and X_j has a 0. This degree of similarity can then be applied to calculating the degree of dissimilarity between two vectors as follows:

$$Dissimilarity\ \left(X_i, X_j\right) = 1 - Similarity\left(X_i, X_j\right) = 1 - \frac{M_{11}}{M_{01} + M_{10} + M_{11}} \tag{16}$$

A new binary solution generator algorithm (NBSG) proposed in [28] is employed for mutation operation and follows three steps.

Step-1: In this step, the degree of dissimilarity between the two vectors is computed using Eq. (16). It is then multiplied with a randomly generated scaling factor 'r' to calculate 'DA,' which is used in the next step of the NBSG algorithm.

$$DA = r.Dissimilarity\left(X_i, X_j\right) \tag{17}$$

The '*r*' is taken within a specified maximum ($r_{\max}$) and minimum ($r_{\min}$), or it can be taken as a fixed decimal number between 0 and 1, as shown in Eq. (17).

$$r = r_{\max} - \frac{(r_{\max} - r_{\min})}{Total\ number\ of\ iterations}(current\ iteration\ number) \tag{18}$$

After the calculation of DA, it is used in a mathematical programming model to find the quantities M_{01}, M_{10}, M_{11} of the new solution vector $V^t_i$by a total enumeration scheme (TE).

The model is described as:

$$\min\left|1 - \frac{M_{11}}{M_{01} + M_{10} + M_{11}} - DA\right| \tag{19}$$

Subject to the constraints as follows.

$$M_{01} + M_{11} = n_1 \tag{20}$$

$$M_{10}\ \leq\ n_0 \tag{21}$$

$$M_{10}, M_{11}\ \geq\ 0 \tag{22}$$

In the TE scheme, all the possible combinations $(n_1\ + 1)(n_0\ + 1)$are evaluated to determine the solution which contains the quantities. M_{01}, M_{10}, $M_{11}$of the new solution. The new solution vector is then initialized as a 1xD zero vector.

Step-2: In the second step of NBSG, any M_{11} number of zero bits from $\mathbf{V_i}$ are selected such that their corresponding positions in $X_i$contains ones and are then changed to ones.

Step-3: In the third step M_{10} number of zero bits of $\mathbf{V_i}$ are randomly picked such that their corresponding positions in $\mathbf{X_i}$It contains zeros and is then changed to ones. Thus, a new solution vector $\mathbf{V_i}$ is generated. All other steps of the DABC algorithm are similar to its continuous version.

Modified discrete version of ABC algorithm

The major problem encountered in the case of the DABC algorithm is that the optimum solution is unreachable even after getting closer to near-optimal values in the search direction and a slower convergence rate in the case of many variables. The reasons that might be responsible for this type of characteristic of DABC are (i) lack of exploration of solutions in any particular direction of search space efficiently even after reaching the direction because of minimum use of the local search step, (2) lack of having an efficient local search step suitable for PMU placement problem, (3) inability to get out of the local minima trapping problem and (4) random replacement of solutions in every iteration. This work proposes three modifications to circumvent the above problems for this Algorithm.

The first and second problems can be overcome by applying a better local search technique which can help to further minimize the number of ones in the solution vector as a result, each search direction explored will be examined efficiently to lead to any feasible but better solution vector. In the local search technique used in the DABC algorithm, the total number of 0 s and 1 s are kept constant in a solution vector even after applying a swap operation to some of the bits. Because of this, only the solution vectors having the same number of PMUs are tested, and no decrease in the number is encountered. Therefore, the first modification in MDABC is to perform a swap operation in the local search step, changing the number of 0 s and 1 s in the solution vector. In this step, anyone bit is randomly chosen and swapped. Because of this, any infeasible solution vector that fails to satisfy the problem constraints can become feasible by placing another PMU. Also, in case of a feasible solution vector having more PMUs might change by removing anyone PMU randomly.

The second modification is about changing the probability of execution of the local search step, which offers mutation through swapping. In the DABC algorithm, it was kept at 0.01. Here the rate of application of local search mutation on the population is obtained using a fuzzy logic method. Generally, the diversity and convergence characteristics of any population are exhibited by the incremental changes in the standard deviation of its fitness distribution ($\Delta\sigma$) and average fitness value (Δf_{avg}). These two parameters determine the mutation probability of the local search phase with a fuzzy rule base. As a result, the rate of execution of efficient local search is revamped consistently in every iteration.

The third and fourth problems can be overcome by occasionally introducing new random solution vectors into the population. In the DABC algorithm, random solution vectors are utilized in every iteration to replace the solutions vector from which no better solution vector had been obtained after mutation steps in each iteration. The third modification proposed is the inclusion of stall limits. Replacement of those solutions is avoided in every iteration. Instead, a limit is applied to the number of mutation steps only. If no better solution is generated, a new random solution vector is inserted for that position in the population. Those solutions are kept in the population for some more time by increasing the number of times the mutation and local search operations are applied to those. Any search direction is examined efficiently to contain any better solution. The probability of getting a better solution from those increases and those parts of the search space is efficiently utilized.

The complete Algorithm for MDABC can be presented in six steps as follows.

Step-1: A population of SN number of employed bees is initialized, representing solution vectors with randomly chosen zeros and ones in different positions of the vectors. Each vector contains a D number of elements taken, similar to a network's total number of buses (N). Fitness values are computed using the objective for each solution vector in the population, and the best value is stored along with the corresponding solution vectors. The iteration number is initialized to 1.

Step-2: Employed bees phase is executed, the new solutions are computed using the NBSG algorithm, and each is evaluated. Those newly generated vectors having better fitness functions generated for any particular position in the population replace the old vectors in the population. With each vector replacement in the population, the corresponding position stall count is updated to 0, otherwise, it is incremented by one.

Step-3(Modified): In the onlooker phase, depending upon the normalized fitness values and a random number p_h, a solution vector from the population is selected for generating a new solution using NBSG. This selection process is used repeatedly for all the individuals in the population. If found, new vectors better replace the old ones in their corresponding population positions. Stall counts are updated after each replacement.

Step-4 (Modified): A local search step selects random vectors from the population several times (L_n) specified at the start. Then with a probability of L_p obtained from applying a fuzzy rule base, a swapping operation is performed on these. It selects any one-bit position in that solution vector and then changes it to 1 if it stores a 0 and vice versa. A fuzzy rule base for two inputs as the standard deviation of fitness distribution ($\Delta\sigma$) and average fitness value (Δf_{avg}) of the population under consideration and mutation probability (L_p) as output are formulated as shown in Table 1. The five input fuzzy sets are considered as LN (Large Negative), SN (Small Negative), ZR (near Zero), SP (Small Positive), and LP (Large Positive). Their corresponding membership functions are shown in Fig. 1. Five fuzzy singletons, i.e., VS(Very Small), S (Small), M (Medium), L (Large), and VL (Very Large)with central values at 0.1, 0.3, 0.5, 0.7, and 0.9 are used respectively by the output of the fuzzy system. The max-min rule for fuzzy inference mechanism and centroid defuzzification are utilized for evaluating the rule base.

Table 1
Fuzzy rule base for Local Search mutation probability.

		LN	SN	ZR	SP	LP
	LN	VL	VL	L	L	M
	SN	VL	L	M	M	M
Δf_{avg}	ZR	L	M	M	S	S
	SP	L	M	S	S	VS
	LP	M	M	S	VS	VS

Step-5(Modified): In the scout phase, the stall counts of each vector position in the population are checked not to exceed a certain stall limit. The solution vector is replaced by a newly initialized vector when any position's stall count exceeds the specified stall limit. At the end of the scouting phase, the iteration number is incremented by 1.

Step-6: Steps-2 to step-5 are repeated for a certain number of cycles. The best solution is selected and stored according to the fitness value at the end of each iteration. After meeting the termination criteria of completing a certain prior set number of iterations, the best solution vector obtained at the last iteration is announced as the optimal solution to the problem.

The modified steps' efficacy can be proved by applying both DABC and MDABC to standard benchmark functions of the uncapacitated facility location (UFL) problem mentioned in [28]. In this work, two benchmark functions, i.e., Cap-71 and Cap-133, are considered and taken from the OR-Library [31]. The results are compared to various performance indices used in [28].

Test cases

Experiment 1:

To examine the convergence rate of both algorithms, two graphs are presented, which are obtained in a typical run of the problems. The MDABC attains the optimum solution with much fewer iterations than the DABC algorithm (Figs. 2 and 3).

Experiment 2:

In another experiment, the following values of control parameters are utilized for the two algorithms. The total number of bees SN is 10, the total number of iterations is 30, and r_{max} (0.9), r_{min} (0.5), and L_n(50) are kept the same for both algorithms. The proposed local search step and stall limits are utilized for the Cap77 and the Cap133 test problem. The results are obtained as an average of over 30 runs and are presented in Table 2 and 3 for Cap71 and Cap133 problems, respectively. Table 4 shows the results of both algorithms after 2000 iterations and with the same number of bees as 30. The two performance indices GAP(%) decreases considerably, and #OPT increases with MDABC.

Three performance indices are taken from [28] for Comparison. Those are GAP(%), #EVL, #OPT. GAP(%) indicates the average gap between (1) the best cost values obtained in each run and (2) the optimal cost value as a percentage of the optimal cost value. The #EVL is the total average number of function evaluations executed for all the trials. The #OPT indicates the total number of times the Algorithm reaches the optimal cost value from all the trials.

The results obtained with each modification of the Algorithm are presented.

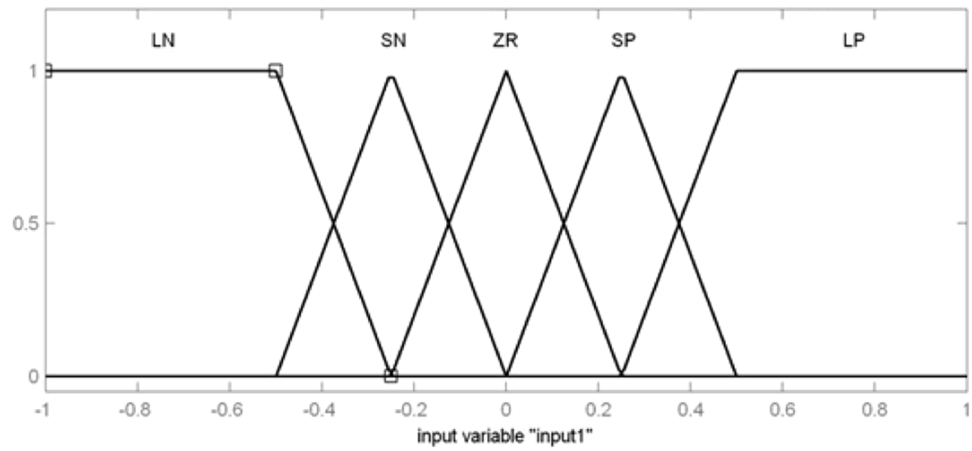

Fig. 1. Membership functions for inputs of the fuzzy system.

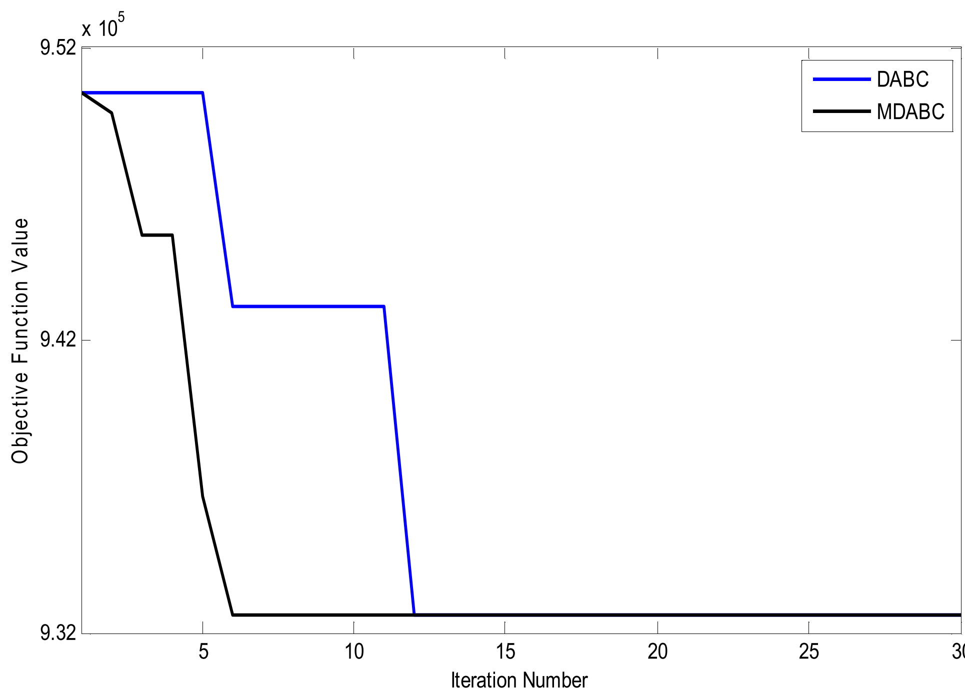

Fig. 2. Comparison results for the Cap71 problem.

1. The first modification shown in Tables II and III includes the stall limit with the new swap technique in the Algorithm by which the GAP(%) gets better.
2. The second modification uses an FL-based mutation probability (Lp) to select individuals from the population for the local search operation. The results indicate that this step changes the number of individuals to which the local search mutation is applied in each iteration. According to the presence of good individuals with better fitness in the population, the GAP(%) decreases further, implying better convergence characteristics.
3. The third modification considers the pH value as random instead of 0.5 or 1. This increases the selection of individuals from the population in the onlooker phase but does not select all the individuals for mutation, as is considered when pH is taken as 1 in DABC. In some cases, this might increase the number of evaluations, but the GAP(%) further reduces.
4. The fourth modification is done by adjusting the maximum stall limit to different values by which the results can be reached in fewer iterations. As compared to all the results mentioned in [28], the proposed Algorithm uses a less number of evaluation steps(EVL) and reaches the optimum (OPT), e.g., the maximum number of times.

Result analysis for the optimal PMU placement case studies

Two major factors compromise the objective function of the PMU OPP problem. The first factor is the number of PMUs to be installed in a network of N buses, which must be minimized simultaneously. The second factor is considered the fulfillment of the complete observability of the network. The fitness function in this study is modified to transform the constrained minimization OPP problem into an unconstrained maximization problem.

The fitness function is calculated as follows,

$$fitness_i = \begin{cases} 1 - \dfrac{\left(\sum_{i=1}^{N} x_i\right)}{N} & \text{if all the buses are observable} \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

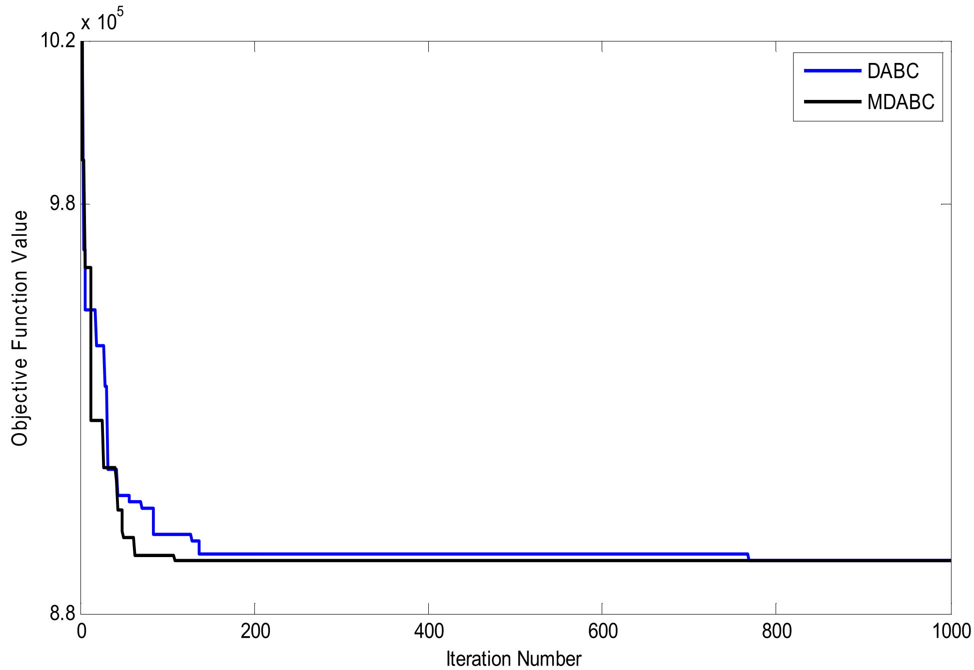

Fig. 3. Comparison results for the Cap133 problem.

Table 2
Comparison results for the Cap71 problem in 30 iterations.

Method Applied	Problem Name	Problem Size	Stall Limit	P_h	L_p	GAP(%)	#EVL	#OPT
DisABC [10]	Cap71	16×50	No	0.5	0.01	0	3192.8	30
MDABC	Cap71	16×50	100	0.5	0.01	9.0660e-005	1.1027e+003	30
MDABC	Cap71	16×50	100	0.5	0.3	0	1066	30
MDABC	Cap71	16×50	100	random	0.3	0	1.0437e+003	30
MDABC	Cap71	16×50	100	Random	0.5	0	1.3569e+003	30

Table 3
Comparison results for the Cap133 problem in 30 iterations.

Method Applied	Problem Name	Problem Size	Stall Limit	P_h	Lp	GAP(%)	#EVL	#OPT
DisABC [10]	Cap133	50×50	No	0.5	0.01	3.27e-2	25,086	18
MDABC	Cap133	50×50	100	0.5	0.01	0.0246	774.3333	3
MDABC	Cap133	50×50	100	0.5	0.3	0.0147	1091	9
MDABC	Cap133	50×50	100	random	0.3	0.0135	1.2097e+003	11
MDABC	Cap133	50×50	50	random	0.3	0.0109	1.2121e+003	24

Table 4
Comparison results for Cap133 problem in 2000 iterations.

Method Applied	Problem Name	Problem Size	Stall Limit	P_h	L_p	GAP(%)	#EVL	#OPT
DisABC [10]	Cap133	50×50	No	0.5	0.01	1.84e-2	53,659.9	23
MDABC	Cap133	50×50	400	0.5	0.3	9.9581e-005	1.5048e+005	30

A solution vector is considered better than another only when it satisfies the system observability and is assigned a non-zero fitness value. All the possible values of the number of PMUs are normalized between 0 and 1.

In this work, the MDABC has been applied to prove its effectiveness in determining the optimal solution to the OPP problem. It is simulated for various test systems such as IEEE −14, −30, −39,−57 bus systems. The line data and the corresponding bus data of various IEEE test systems are obtained from [32–33]. Table 5, 7, 9, 11, 13, and 15 shows the results of the OPP for test systems under different system constraint circumstances. Each result in this work matches the minimum number obtained in other works, as seen in Tables 6, 8, 10, 12, 14, 16. It proves

Table 5
Optimal PMU placement results under Case 1 conditions.

Test systems	PMUs number	Optimal locations under normal operating conditions
IEEE 14 bus	4	2, 6, 7, 9
IEEE 30 bus	10	1, 7, 9, 10, 12, 15, 18, 25, 28, 30
IEEE 39 bus	13	2, 6, 9, 10, 13, 14, 17, 19, 20, 22, 23, 25, 29
IEEE 57 bus	17	3, 6, 10, 19, 22, 25, 27, 32, 36, 39, 41, 45, 46, 49, 52, 55

Table 6
Comparison of the optimal number of PMUs resulted in Case 1.

Test Systems	Proposed	Ref [33]	Ref [34]	Ref [35]
IEEE 14 bus	4	4	4	4
IEEE 30 bus	10	10	–	10
IEEE 39 bus	13	–	–	13
IEEE 57 bus	17	17	17	–

that a binary non-classical optimization technique can determine the solution to the OPP problem.

Case-1: Complete topological observability without consideration of zero injection effect

The PMU placement set under normal operating conditions without considering the effect of ZIBs is determined and is presented in Table 5. The observability analysis procedure mentioned in Section 3 is carried out with the exclusion of Step 3. Table 6 presents the comparative results obtained based on the number of PMUs for this case using the proposed MDABC and other techniques, such as integer linear programming methods.

Case-2: Complete topological observability with consideration of the ZI effect

When the effect of ZIBs is included in the observability analysis technique, it considerably reduces PMU numbers in the optimal placement set while ensuring system observability criteria. This proves to be highly cost-effective. The fitness function, in this case, is evaluated using Eq. (23). Table 7 presents the results under this condition and is compared with other works in Table 8. In this case, using the proposed method, the results are better for the 57-bus system. Also, with proper adjustment of MDABC algorithm parameters, the problem of local minima trapping, which is more evident in the 57 bus system and has an effect of settling to 12 PMUs instead the minimum of 11 PMUs, can be easily overcome. A minimum 11 number of PMUs is obtained in every run of MDABC.

Case-3: Complete topological observability under a single PMU outage with consideration of the ZI effect

A measurement system design must withstand the failures of various measuring devices and should be able to provide all the measurement information even in case of any PMU failure. This can only be ensured by getting each bus observable at least two times by a robust PMU placement scheme. When observability is tested with the zero injection bus effect consideration, each bus phasor can be estimated by either a PMU at that bus or on any of its adjacent buses through direct or indirect measurement or by solving a set of KCL equations of the ZI buses. But if the determined optimal placement set is such that a measurement provided by a PMU placed at a bus is used for determining the bus voltage phasor of one of its adjacent buses, which are involved in any of the ZIB group KCL equations for getting its second time observable, failure of the corresponding PMU will lead to unobservability of that bus. To avoid such problems, any solution vector is tested to ensure complete observability under the failures of all the PMUs in the solution vector considered one at a time. If a solution vector satisfies this constraint, the correct solution is declared, and the optimization problem is carried out with the corresponding fitness function.

The fitness function for each solution vector set is evaluated as follows,

$$fitness_i = \begin{cases} 1 - \dfrac{\left(\sum_{i=1}^{N} x_i\right)}{N} & if\,all\,the\,buses\,are\,observable\,for\,each\,PMU\,failure \\ 0 & otherwise \end{cases} \tag{24}$$

Table 9 presents the PMU placement results of various test systems determined by the proposed method for this case. These are compared with the other results presented in Table 10.

Case-4: Complete topological observability under a single line outage with consideration of the ZI effect

A line outage situation can be simulated by removing the corresponding line from the set of lines. The system connectivity matrix must be accordingly modified to reflect the effect. For getting a placement setting that must provide measurement data even under any single line outage situation, the observability analysis procedure should be carried out under all the line outage cases considering one at a time. Each line is simulated to be removed, and the observability of the corresponding terminal buses is examined in the procedure. The fitness function for each solution vector is evaluated as follows,

$$fitness_i = \begin{cases} 1 - \dfrac{\left(\sum_{i=1}^{N} x_i\right)}{N} & if\,all\,the\,buses\,are\,obseravle\,for\,each\,line\,outage \\ 0 & otherwise \end{cases} \tag{25}$$

Tables 11 and 12 present the PMU placement results of various test systems for line outage contingency conditions and the comparison with

Table 7
Optimal PMU placement locations result for Case 2 conditions.

Test systems	No. of Zero Injection buses	Locations of Zero injection buses	No. of PMUs	Optimal locations under normal operating conditions considering ZI buses
IEEE 14 bus	1	7	3	2, 6, 9
IEEE 30 bus	6	6, 9, 22, 25, 27, 28	7	3, 5, 10, 12, 19, 24, 30
IEEE 39 bus	12	1, 2, 5, 6, 9, 10, 11, 13, 14, 17, 19, 22	8	3, 8, 12, 16, 20, 23, 25, 29
IEEE 57 bus	15	4, 7, 11, 21, 22, 24, 26, 34, 36, 37, 39, 40, 45, 46, 48	11	1, 6, 13, 19, 25, 29, 32, 38, 42, 51, 54

Table 8
Comparison of the optimal PMUs number found using different techniques for Case 2.

Test Systems	Proposed	Ref [33]	Ref [36]	Ref [36]
IEEE 14 bus	3	3	3	3
IEEE 30 bus	7	7	7	7
IEEE 39 bus	8	–	–	8
IEEE 57 bus	11	12	11	–

Table 9
Optimal PMU placement locations result for Case 3 conditions.

Test systems	PMUs number	Optimal locations result in a single PMU outage
IEEE 14 bus	7	1, 2, 4, 6, 9, 11, 13
IEEE 30 bus	15	2, 3, 4, 7, 10, 12, 13, 15, 17, 19, 20, 21, 25, 27, 30
IEEE 39 bus	18	4, 7, 8, 10, 16, 18, 20, 21, 23, 25, 26, 29, 32, 34, 36, 37, 38, 39
IEEE 57 bus	26	1, 2, 4, 7, 12, 13, 18, 19, 25, 27, 29, 31, 32, 33, 35, 38, 41, 42, 45, 47, 48, 50, 51, 53, 54

Table 10
Comparison of the optimal PMUs number for Case 3 conditions.

Test Systems	Proposed	Ref [33]	Ref [37]
IEEE 14 bus	7	7	7
IEEE 30 bus	15	17	15
IEEE 39 bus	18	–	18
IEEE 57 bus	26	26	26

Table 11
Optimal PMU placement locations result for case 4 conditions.

Test systems	PMUs number	Optimal locations result in a single-line outage
IEEE 14 bus	7	1, 3, 6, 8, 10, 13, 14
IEEE 30 bus	13	1, 3, 5, 10, 11, 13, 14, 15, 16, 19, 23, 26, 29
IEEE 39 bus	15	3, 8, 16, 23, 27, 28, 30, 31, 32, 33, 34, 35, 36, 37, 38
IEEE 57 bus	19	1, 2, 6, 12, 14, 19, 27, 29, 30, 32, 33, 38, 41, 45, 50, 51, 53, 55, 56

Table 12
Comparison of the optimal PMUs number for Case 4 conditions.

Test Systems	Proposed	Ref [36]	Ref [37]
IEEE 14 bus	7	7	7
IEEE 30 bus	13	13	13
IEEE 39 bus	15	15	15
IEEE 57 bus	19	19	19

other works, respectively.

Case-5: Complete topological observability under a single PMU or line outage with consideration of the ZI effect

In this case, the placement setting is determined to anticipate the failure of any contingency that occurred one at a time. The observability analysis is carried out by simulating each PMU failure or each line failure and calculating the number of observable buses. The solution that provides observability under any single line outage or single PMU outage from the solution set is announced as optimal.

$$fitness_i = \begin{cases} 1 - \dfrac{\sum_{i=1}^{N} x_i}{N} & if\, all\, the\, buses\, are\, observable\, for\, each\, line\, or\, PMU\, outage \\ 0 & otherwise \end{cases} \tag{26}$$

In this case, the PMU placement results are presented in Table 13 and then compared with other results in Table 14.

The PMU number attained using the proposed technique is better than that of [37]. In some of the works, a lower number of PMUs have been attained compared to the proposed approach or the same minimum

Table 13
Optimal PMU placement locations result for Case 5 conditions.

Test systems	PMUs number	Optimal locations result under a single line outage or single PMU outage condition.
IEEE 14 bus	8	1, 2, 4, 6, 8, 9, 11, 13
IEEE 30 bus	17	2, 3, 4, 5, 9, 10, 11, 12, 13, 15, 17, 18, 20, 23, 26, 27, 30
IEEE 39 bus	22	2, 4, 7, 9, 12, 16, 18, 20, 23, 25, 26, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39
IEEE 57 bus	26	1, 2, 6, 9, 12, 15, 18, 19, 21, 26, 27, 29, 30, 31, 32, 33, 35, 38, 41, 46, 50, 51, 52, 54, 55, 56

Table 14
Comparison of the optimal PMUs number for Case 5 conditions.

Test Systems	Proposed	Ref [37]
IEEE 14 bus	8	8
IEEE 30 bus	17	17
IEEE 39 bus	22	22
IEEE 57 bus	26	26

number of PMUs by incorporating different preplaced conventional measurements in the measurement system with PMUs. But including those in the OPP problem can result in a placement set that can lead to a nonlinear state estimator. This is because of the different data rates of PMUs and those devices, more complicated estimation, and more margins of error. The full observability condition cannot be maintained anymore independent of the existing measurement system. Therefore the incorporation of the conventional measurement is avoided in the OPP problem formulation in this work.

Case-6: Complete observability under the placement of branch PMUs

Table 15 shows the results of an optimum number of branch PMUs with their locations and channel assignment strategy under the normal operating condition with new constraints included in the problem formulation. Those are compared with the results of [38] and [39] in Table 16. The inconvenience encountered in [39] due to the placement of multiple numbers of branch PMUs at the same bus location can be avoided in this work with the same number of minimum PMUs placement nevertheless at different buses. The PMUs are distributed across the whole network instead of placing more than one PMU on the same bus. The theoretical minimum number of PMUs ((N-1)/2)+1, can be attained using MDABC for different N bus test systems.

Comparison of results of DABC and MDABC

A comparison is presented to show the incapability of the DABC algorithm to solve the PMU placement problem. The DABC algorithm in [28], when used for the PMU placement problem, failed to give the optimum results for different operating conditions. The results of the DABC, BPSO, and MDABC algorithm is shown in Fig. 4. These algorithms are applied under normal operating conditions for a 57-bus test system. The results are obtained for 5000 iterations with a population size of 200 for DABC, 200 for BPSO, and 20 for MDABC. It can be seen that although BPSO and MDABC reach the optimal solution, DABC fails to attain the

Table 15
Branch PMU placement locations results under Case 6 problem condition.

Test systems	PMUs number	Branch PMUs positions	Measurements assigned to branch PMUs
IEEE 14 bus	7	2, 5, 6, 7, 9, 11, 13	2–3, 1–5, 6 –12,7–8,4–9,10–11,13–14
IEEE 30 bus	15	1, 4, 7, 8, 9, 10, 13, 15, 16, 19, 22, 23, 26, 28, 30	1–3,2–4,5–7,6–8, 9–11,10–20,12–13,14–15, 16–17,18–19,22–24,21–23,25–26, 27–28, 29–30
IEEE 39 bus	21	2, 3, 4, 8, 9, 10, 11, 15, 19, 20, 21, 22, 23, 25,26, 27, 30, 31, 32, 36, 38	1- 2, 3–18, 4–5, 7–8, 9–39, 10–13, 11–12, 14–15, 19–33, 20–34, 16–21, 22–35, 23–24, 25–37, 26–28, 17–27, 2–30, 6–31, 10–32, 23–36, 29–38
IEEE 57 bus	29	2, 5, 7, 9, 11, 12, 13, 14, 16, 18, 19, 21, 23, 25, 26, 28, 30, 32, 34, 36, 40, 42, 44, 45, 48	2–3, 5–6, 7–8, 9–10, 11–43,12–17,13–49,14- 46, 1–16, 4–18,19–20, 20–21,22–23,24–25, 26–27, 28–29, 30–31, 32–33, 34–35, 36–37,40–56,41–42, 38–44, 15–45, 47-48,50–51, 52–53,54–55, 39–57

Table 16
Comparison of the optimal branch PMU placement results.

Test Systems	Proposed	Ref [38]	Ref [39]
IEEE 14 bus	7	–	7
IEEE 30 bus	15	15	15
IEEE 39 bus	21	–	–
IEEE 57 bus	29	–	29

minimum PMUs number even at the end of 5000 iterations. Also, a slower convergence characteristic is encountered in the case of DABC, as shown in Fig. 4. On the contrary, MDABC can provide a result within 500 iterations compared to BPSO within 2500 iterations.

Comparison of results of BPSO and MDABC

A comparison is carried out to see the convergence rate of BPSO and MDABC algorithms in 200 iterations, and the graph is shown in Fig. 5 for Case 2 conditioned OPP problem in the case of 57 bus systems. The graph shows that the minimum number of PMUs is computed much before the proposed approach compared to BPSO. Therefore, MDABC is more suitable for application in this problem than other heuristic techniques.

An experiment was conducted with the same population size of 200 and the same number of iterations (1000) in both BPSO and MDABC for the case 2 condition simulations. From results obtained after 30 runs in both the programs, the best value, the worst value, and the average value of the minimum PMUs number are calculated and shown in Table 17.

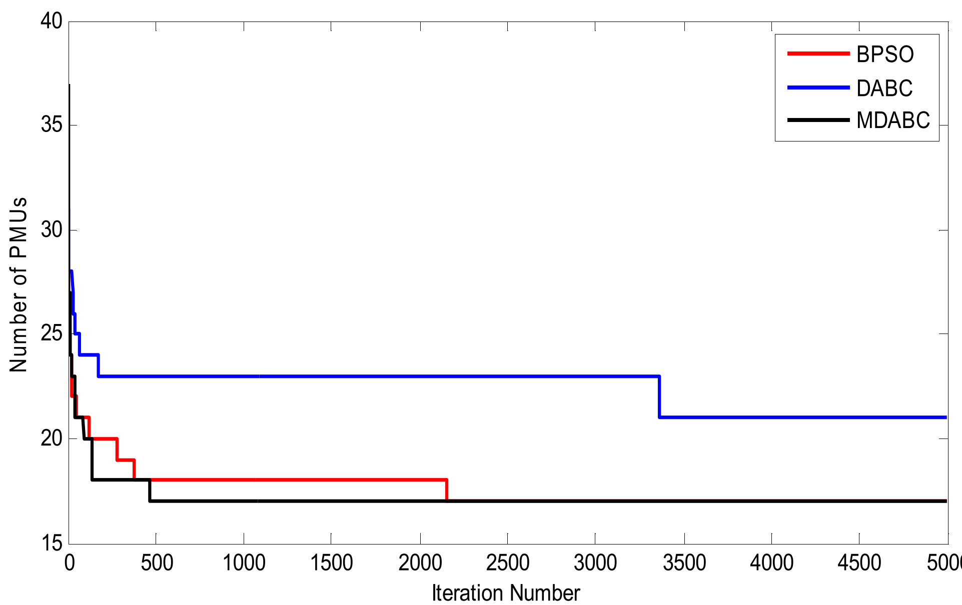

Fig. 4. Comparison between the results of BPSO, DABC, and MDABC for the OPP problem.

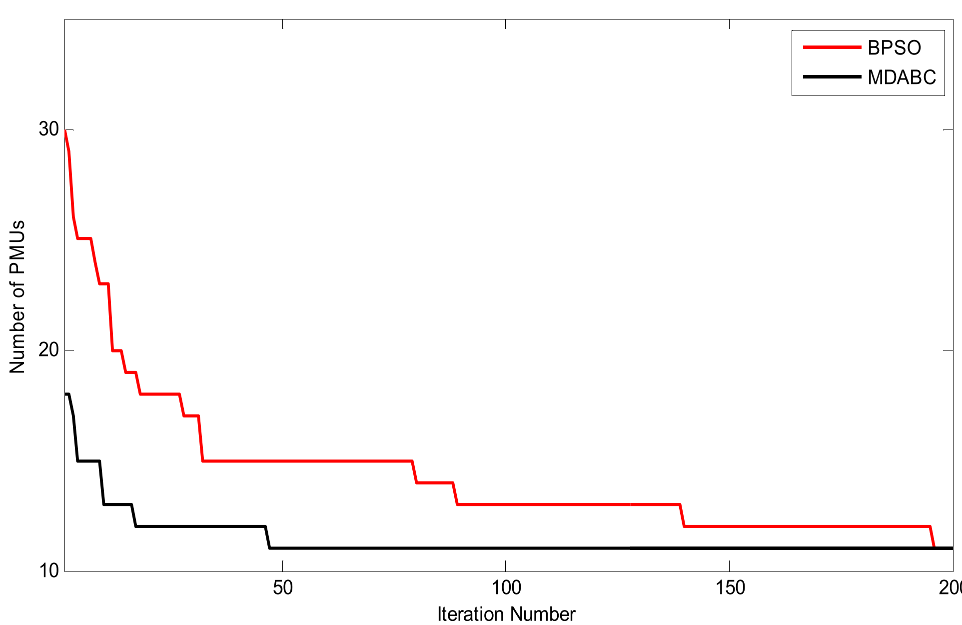

Fig. 5. Results of BPSO and MDABC for Case 2 conditions.

Table 17
Comparison of the results from DisABC and BPSO for the IEEE 57 bus test system.

Technique applied	Best value of the minimum PMU number	The average value of the minimum PMU number	Worst value of the minimum PMU number
BPSO	11	11.96	12
MDABC	11	11.13	12

Critical analysis and future scope of research

Looking at the recently done research outcomes and the future trends of the Smart Grid scenario, the following scope can be presented for the problem related to PMU's optimal location.

- As discussed in the review in the introduction section, all the methods ranging from classical to non-classical techniques, have limitations, and it is hard to confirm the best out of all the suggested methods in recent times. However, in many engineering problems, hybrid approaches are very successful in providing an optimal solution due to reducing the individual limitations as taken in this proposed approach. So, the hybrid approach related to heuristic algorithms with linear and nonlinear programming may be a better choice in the future.
- In the OPP problem, multiple optimal sets have resulted in all full observability. It is essential to formulate the index to choose the best one out of many optimal solutions.
- Less work has been done on OPP considering multi-objective formulation considering many desired factors, including installation cost, redundancy, performance, and other planning constraints. This provides more feasible and realistic practical results.
- Many constraints are taken to date on the OPP formulation, like the effects of the ZI bus, the effect of conventional measurements, single and multiple PMU loss contingencies, single branch outage contingencies, and the effect of PMU channel capacity extensively. However, there is a need to consider additional constraints looking from the operational perspective, such as the influence of PMU malfunction, communication failures, measurement redundancy, metering accuracy, and uncertainty.
- Generally, the OPP is formulated extensively by considering integer and binary placement decision variables. It is possible to formulate the OPP problem considering the decision variables as continuous and relaxing the constraints.
- Recently Graph Theory and Graph-based algorithms have been merged as advanced computational methods used in many engineering problems and can be considered an alternative formulation of the OPP problem. This can be considered an interesting area of research in the direction of optimal PMU locations.
- All the approaches are simulated and tested for small-scale networks. Large-scale networks case the OPP problem, and its solution brings many computation-related issues like computational time, complexity in the formulation, and the possibility of being trapped at local minima. So, the robustness of all the methods should be investigated further, and their reliability in providing optimal solutions needs to be tested and analyzed.
- The generalization of the PMUs reliability model needs to be investigated further for better pragmatic analysis considering multiple parameters, including data uncertainties and loss, single and multiple fault patterns, and inaccuracy in measurement. For the reliability evaluation of optimal PMU placement, Bayesian networks, Petri nets, and the Markov model can be considered for future reliability evaluation while deciding the OPP set of the last decisive buses.

Conclusion

The proposed FL-DABC optimization technique effectively computes the minimal number of PMUs required for full observability under various power system contingencies. This robust placement set accounts for real-time operational reliability. The FL-DABC method also calculates branch PMU placement in strictly radial networks, incorporating a constraint to prevent multiple PMUs on a single bus. Though time-intensive for large systems, heuristic techniques offer faster convergence and fewer computations than classical methods, such as integer programming or exhaustive search. While classical and heuristic approaches may yield multiple solutions, the proposed heuristic technique explores the entire search space, enabling the selection of the best solution based on installation criteria. Furthermore, fuzzy mutation enhances the search process with reduced computational time.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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